DIGITAL TRUST FUTURES
FOUNDATION

DIGITAL SOVEREIGNTY IN PRACTICE

Sovereignty by Design

Building DPI architecture, roadmaps and regional resilience

Digital Sovereignty Series | Article 4 of 6

Why this matters now

Sovereignty cannot be bolted onto Digital Public Infrastructure after launch without cost, disruption and compromise. This article turns the policy debate into an architecture and implementation question, showing how countries can design sovereignty into DPI from the beginning and build it progressively through national and regional capability.

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Executive summary

Digital Public Infrastructure is where digital sovereignty becomes a practical design problem. A national identity system, payment rail or data exchange layer is sovereign in practice only if its architecture, operations, governance, data controls and assurance model make control real before the system becomes nationally dependent. This article sets out a six-layer architecture for sovereignty-aware DPI, a realistic phased roadmap, and the role of regional cooperation in giving smaller and lower-capacity countries a credible route to meaningful control.

Key points for policymakers and DPI leaders

- Security and sovereignty controls are cheaper and more effective when designed in before go-live.
- DPI sovereignty requires governance, trust, data protection, platform security, operations and ecosystem controls to work together.
- A credible roadmap is a 36-month-plus commitment, not a procurement milestone.
- Regional infrastructure, shared CSIRT capacity and pooled assurance can turn sovereignty from aspiration into capability.

The central argument is that sovereignty is not a hosting decision or a contractual promise. It is the cumulative result of choices about key custody, administrative access, logging, code control, independent assurance and exit options. When countries leave those choices to vendors or defer them until after launch, they inherit dependencies that are costly to unwind and hard to see.

Progress is therefore best measured in verified capability rather than declared intent. The roadmap in this article moves from immediate visibility of dependencies, through domestic operational control and independent assurance, to long-term regional resilience. For countries that cannot build every capability alone, regional cooperation offers a practical way to share cost and expertise while retaining national accountability for laws, data protection and security thresholds.

Key takeaways

For policymakers, this means writing sovereignty requirements into legal mandates, procurement terms and funding conditions before signing contracts. For DPI leaders, it means building the domestic skills, operating procedures and assurance relationships that turn those requirements into everyday practice. Neither group can deliver this alone, so sustained coordination among policy, technical and regional actors is essential.

1. Treat sovereignty as an architectural requirement from day one, because every control deferred past go-live becomes harder, costlier and less complete to retrofit.
2. Anchor control in the trust layer, since whoever holds the keys, administers the environment and sees the logs ultimately determines whether data sovereignty is real.
3. Measure progress through independent assessment rather than policy statements, so that the gap between declared and actual sovereignty narrows instead of widening over time.

Sovereignty by design is ultimately a discipline of sustained, verifiable choices rather than a single procurement or policy event. Countries that embed control early, build domestic and regional capability in deliberate phases, and test their assumptions independently will be far better placed to protect citizens and sustain trust in national systems. The sections that follow set out how that discipline translates into architecture, sequencing and regional cooperation.

1. Sovereignty must be designed in at the architecture stage

Sovereignty controls retrofitted after national systems go live are usually more expensive, more disruptive and less complete than controls embedded at design time. Once identity, payment, social protection or data exchange systems are widely integrated, later controls must work around existing dependencies, data flows, operational habits and vendor relationships.

2. A six-layer sovereignty-aware DPI architecture

A complete DPI sovereignty posture is not a single product. It is an architecture of mutually reinforcing controls. The six layers below show where sovereignty decisions are made, and where false confidence usually enters if one layer is treated as sufficient on its own.

Layer	What it covers	Sovereignty relevance
Governance	Legal basis, accountability, oversight, procurement controls, incident authority, audit rights.	Determines who actually owns the decisions, not just who operates the system day-to-day.
Trust	Identity, authentication, authorisation, consent, signing, key management.	The layer where data sovereignty is technically enforced or abandoned, depending on who holds the keys.
Data protection	Minimisation, purpose limitation, retention controls, encryption, privacy impact assessment.	Reduces the value and legal exposure of a breach, and underpins protection of personal information.
Platform security	Secure SDLC, API security, secrets management, logging, patching, vulnerability management.	Where supply chain risk inherited from digital public goods enters the system.
Operations	Monitoring, SOC workflows, incident response, backup and recovery, continuity, and third-party risk.	Determines whether operational sovereignty is real, or only a policy aspiration.
Ecosystem	Certification, onboarding controls, interoperability testing, fraud controls, provider offboarding.	The weakest participant in an interoperable network sets the practical security floor for everyone connected to it.

Designing sovereignty in means asking hard questions early: who holds keys, who can administer the environment, who can see logs, who can change code, who verifies security, what happens if the primary vendor is unavailable, how cross-border access is controlled, and how citizens are protected

when systems fail.

The key design principle is **federation over unnecessary centralisation**. Monolithic DPI designs can concentrate risk and create single points of failure. Modular, federated and segmented architectures reduce lateral movement, limit blast radius, and allow individual components to fail or be replaced without collapsing the wider national system.

2.1 The six layers in detail

The six layers operate as one sovereignty control system. Each layer answers a different question about who can decide, who can technically exercise control, and whether that control remains effective during normal operations, disruption, vendor failure or cross-border dependence.

01. Governance

Defines the legal and institutional authority for the DPI: mandate, ownership, accountability, oversight, procurement, audit rights, incident authority, and the rules governing outsourcing or shared infrastructure.

Sovereignty focus. Decision rights must remain clear even when technology and operations are delegated. Contracts, laws and governance bodies should preserve the state's ability to direct, inspect, challenge, replace and recover critical services.

Evidence of control. Named accountable owners; enforceable audit and access rights; incident decision authority; supplier exit and transition clauses; documented oversight and independent assurance.

02. Trust

Controls the roots of digital trust: identity proofing, authentication, authorisation, consent, digital signing and cryptographic key management that determine who or what is trusted across the ecosystem.

Sovereignty focus. Control of keys, privileged identities and trust anchors determines whether technical sovereignty is real. Key generation, custody, rotation, recovery and revocation should not depend on an actor the country cannot independently govern.

Evidence of control. HSM-backed key custody; dual control and separation of duties; controlled privileged access; auditable trust-anchor changes; tested emergency revocation and recovery.

03. Data protection

Governs how personal and sensitive data is collected, used, shared, retained, encrypted and deleted, including minimisation, purpose limitation, residency, cross-border transfers and privacy impact assessment.

Sovereignty focus. Sovereignty requires control over data throughout its lifecycle, not only where a database is hosted. Data flows, backups, replicas, support access and lawful cross-border disclosure pathways must be visible and governed.

Evidence of control. Data inventory and flow maps; classification and retention rules; encryption with controlled keys; privacy impact assessments; deletion controls; monitored cross-border and administrative access.

04. Platform security

Protects the software, APIs, infrastructure and supply chain that make DPI function, including secure SDLC, secrets, logging, patching, vulnerability management and dependency assurance for digital public goods and open-source components.

Sovereignty focus. A country cannot claim meaningful control if it cannot verify what code is deployed, how it was built, what dependencies it contains, who can change it, and how vulnerabilities are found and remediated.

Evidence of control. Controlled source and CI/CD access; SBOM and provenance; signed build artefacts; API and secrets controls; vulnerability and patch evidence; independent code and architecture assurance.

05. Operations

Turns design into sustained capability through monitoring, SOC workflows, incident response, backup and recovery, continuity, administrative procedures, skills and third-party operational risk management.

Sovereignty focus. Operational sovereignty is demonstrated when essential services can be monitored, defended, restored and safely operated even if a primary supplier, cloud service or external support team is unavailable.

Evidence of control. Domestic operating capability; independent telemetry and logs; tested incident runbooks; offline and degraded-mode procedures; recovery exercises; privileged-access oversight; continuity and supplier-failure tests.

06. Ecosystem

Extends sovereignty beyond the core platform to every connected participant through onboarding, certification, interoperability testing, fraud controls, interface requirements, continuous assurance and provider offboarding.

Sovereignty focus. Interoperability distributes risk. The practical security floor is therefore set by the weakest connected participant or interface, making common minimum controls and enforceable participation rules essential.

Evidence of control. Common security baseline; pre-production interoperability tests; certification and periodic reassessment; fraud monitoring; dependency mapping; suspension and offboarding processes; evidence-sharing obligations.

> **The design test**
>
> For each layer, leaders should be able to identify the accountable authority, the technical control point, the evidence that proves control, and the recovery or exit path if a dependency fails. A gap in any one of these four elements can undermine the sovereignty of the whole DPI ecosystem.

Effective sovereignty depends on these layers reinforcing one another, because strong control in one area cannot compensate for critical dependency or weakness in another.

The objective is therefore not national self-sufficiency in every technology, but verifiable authority, meaningful technical control, operational resilience and credible alternatives when dependencies fail or become unacceptable. When these conditions are designed, tested and independently assured across

all six layers, DPI becomes more resilient, accountable and capable of sustaining national control while still benefiting from regional cooperation and global technology ecosystems.

3. A phased sovereignty roadmap

Digital sovereignty is built progressively, not delivered as a single technology programme or procurement outcome. Countries typically begin with a mixture of domestic capability, foreign technology, external expertise, legacy platforms and regional dependencies. The practical objective is therefore to understand those dependencies and deliberately increase the areas where national authorities can govern, operate, verify, recover and make independent decisions over time.

The sequencing matters. Trying to localise every component immediately can increase cost and complexity without necessarily improving control, while delaying critical capabilities such as key custody, operational visibility, independent assurance and recovery can create dependencies that become difficult to reverse. No country achieves full sovereignty at go-live, so a credible plan should distinguish immediate visibility, short-term operational capability, medium-term assurance and long-term resilience.

Timeframe	Priority actions	Sovereignty dimension
Immediate (0–6 months)	Conduct an architecture review to identify foreign single-vendor dependencies. Establish a data residency policy with a technical, not merely legal, enforcement plan. Define minimum domestic operational capability requirements. Inventory any AI components already embedded in deployed systems.	Architectural, Data
Short-term (6–18 months)	Build a domestic DPI operations team. Implement HSM-backed key management with domestic key custody. Establish tested offline operational procedures. Commission an independent security assessment.	Operational, Security
Medium-term (18–36 months)	Achieve software bill of materials (SBOM) compliance for all deployed open-source components. Establish a national CERT/CSIRT with a DPI-specific mandate. Develop regional data sovereignty agreements with neighbouring states. Establish AI governance policy for DPI-embedded AI.	Security, Governance
Long-term (36+ months)	Participate in regional shared infrastructure for resilience. Achieve a mature, independently verified DPI security maturity level. Establish domestic DPI security research capability. Contribute to international DPI security standards.	All five pillars

> **No country achieves sovereignty at go-live**
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> The roadmap above is a 36-month-plus commitment requiring sustained political will, continuous investment in domestic capability, and the institutional honesty to commission independent assessments that may surface uncomfortable findings.

> Countries that treat sovereignty as a procurement outcome rather than an operational discipline will find the gap between declared and actual sovereignty widening over time, not closing.

The roadmap should be treated as a capability-building journey rather than a fixed implementation schedule. Measure progress by demonstrable increases in control, resilience, domestic competence, independent assurance and credible exit options, with each phase reducing dependencies that could materially constrain national decision-making or the continuity of essential digital services.

4. Regional and shared sovereignty architecture

For many smaller and lower-capacity countries, the most realistic path to meaningful digital sovereignty is regional rather than purely national. Building every critical capability independently can be prohibitively expensive, difficult to staff and inefficient to sustain. Shared infrastructure, pooled cryptographic and assurance capability, common procurement standards, regional threat intelligence and shared Computer Security Incident Response Team capacity can give participating countries access to capabilities that would otherwise be out of reach.

Regional sovereignty should therefore be understood as **shared capability without surrendered accountability**. National authorities should retain responsibility for laws, citizen rights, data protection requirements, security thresholds and critical decision-making, while selected technical and operational capabilities can be jointly delivered where this improves resilience, affordability and bargaining power. The objective is not to replace national sovereignty with regional control, but to strengthen national authority through trusted shared mechanisms, transparent governance, enforceable participation rules and independently verifiable safeguards.

The design challenge is consequently not a binary choice between “national” and “foreign” infrastructure. A stronger model combines domestic authority, regional capability, open standards, interoperable architecture and carefully governed use of global technology, so that no single government, supplier, cloud provider or regional institution becomes an uncontested point of control. The central test is whether each country can understand its dependencies, influence the shared environment, independently verify its protections, and maintain continuity or exit if the arrangement no longer meets national requirements.

Key issue	Regional sovereignty consideration	Practical design requirement
1. Governance and decision rights	Shared services can create uncertainty over who has authority during incidents, disputes, upgrades or emergency interventions.	Establish formal regional governance, clearly defined national decision rights, escalation procedures, voting or approval mechanisms, audit rights, and transparent accountability for shared infrastructure.
2. Data jurisdiction and cross-border access	Regional platforms may process, replicate or back up sensitive information across several jurisdictions with different legal and privacy requirements.	Define agreed data classifications, residency and transfer rules, permitted administrative access, cross-border disclosure procedures, logging requirements, and mechanisms for resolving conflicts between national laws.

Key issue	Regional sovereignty consideration	Practical design requirement
3. Cryptographic trust and key custody	Centralising trust anchors or cryptographic keys can create a powerful regional point of dependency or compromise.	Use distributed or pooled key-management models with strong separation of duties, HSM-backed protection, national participation in key governance, dual control, auditable ceremonies, and tested revocation and recovery procedures.
4. Operational resilience and incident response	Smaller states may lack sufficient specialist capacity to monitor, investigate and recover from sophisticated attacks independently.	Establish shared SOC or CSIRT capabilities, common incident taxonomies, regional threat-intelligence exchange, coordinated exercises, minimum national response capability, and clear authority for cross-border incident coordination.
5. Supplier and technology dependency	Regional procurement can reduce cost but can also concentrate dependence on a single technology provider or architecture.	Use open standards, modular architectures, common exit clauses, source and configuration portability, interoperable alternatives, independent security assurance, and procurement structures that preserve switching and substitution options.

A mature regional model should be judged not by how much infrastructure is shared, but by whether sharing increases each participating country's practical control and resilience. Regional arrangements should make countries less vulnerable to single-vendor dependence, skills shortages, fragmented security capability and unequal negotiating power, while preserving the ability of national authorities to enforce their own legal and policy requirements.

1. **Share capability, not accountability.** Regional infrastructure can be jointly operated, but responsibility for citizen protection, national law and sovereign decision-making should remain clearly assigned.
2. **Design for reversibility.** Every shared arrangement should include interoperability, portability, recovery, substitution and exit mechanisms so cooperation does not become irreversible dependency.
3. **Verify regional trust continuously.** Common controls, independent assurance, joint exercises and transparent evidence should demonstrate that shared sovereignty arrangements remain effective as technology, suppliers, risks and political relationships change.

Regional sovereignty works best when cooperation strengthens, rather than dilutes, national control. The goal is to create shared capability that expands resilience, technical depth and strategic choice while preserving clear accountability at country level. Done well, regional architecture turns sovereignty from an expensive aspiration into a practical, scalable and independently verifiable operating model.

> **Series bridge**
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> The next article addresses the most common implementation trap in this area: assuming that data is sovereign because it is hosted locally. It connects data protection law, protection of personal information, and the risk of extraterritorial access.

About the author

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Godfrey Kutumela is a globally recognised cybersecurity practitioner with over 26 years of experience shaping secure digital ecosystems across fintech, identity and public infrastructure environments. He operates at the intersection of Digital Public Infrastructure (DPI), Digital Public Goods (DPGs) and national-scale platforms, bringing deep expertise in securing systems where the impact of failure is measured at the population level rather than the enterprise level.

- Trusted advisor to governments, foundations and ecosystem builders delivering secure-by-design DPI across Africa, Latin America and global markets.
- Core contributor to large-scale platforms including Mowali (mobile money interoperability infrastructure), Mojaloop (interoperable faster payments), MOSIP (digital identity) and Tazama (fraud systems).
- Designs and embeds independent, risk-based security practices across payment systems, identity frameworks and cross-border financial infrastructure.
- Works across policy, architecture and implementation layers, translating global standards into deployable security controls.
- Contributor to the Digital Public Goods Alliance (DPGA) Privacy Expert Group, shaping global privacy and security indicators.
- Active in open-source security, DevSecOps integration and supply chain assurance for reusable digital platforms.
- Delivered independent security reviews, penetration testing and architecture governance across critical DPI implementations.

His leadership philosophy in the DPI ecosystem is clear: security forms the foundation upon which safety and trust are built. Safety is what citizens experience in practice, and trust is the legitimacy that sustains adoption; once lost, it is difficult to restore.

About Digital Trust Futures Foundation

Digital Trust Futures Foundation is an independent, non-profit public-benefit organisation strengthening the safety, security, rights, inclusion and trustworthiness of digital public infrastructure (DPI), digital public goods (DPGs), open-source technology and AI-enabled public services. Its work is grounded in the recognition that DPI has become critical national infrastructure operating in a persistent threat environment, requiring independent, ecosystem-level assurance that goes beyond traditional approaches.

The Foundation operates outside vendor, implementer and platform incentives. It combines evidence-based cybersecurity, safety and AI-governance assurance with human-rights and child-protection expertise, and releases its tools, research and frameworks as reusable public goods rather than proprietary products — so that the value of a single engagement extends to every government, maintainer and community facing the same risk.

Digital Trust Futures Foundation NPC is registered as a non-profit company in South Africa and works across Africa and the wider Global South, with emphasis on countries and communities adopting DPI under capacity and assurance constraints.

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References and further reading

This list is organised by institutional category to help readers navigate quickly to the type of source most relevant to their work. It reflects the regulatory and threat landscape as understood at the time of writing, and should be revisited periodically: several of the instruments listed remain subject to active negotiation and change.

United Nations and UN agencies

- UN Office for Digital and Emerging Technologies (ODET) — Universal DPI Safeguards Framework and DPI Safeguards Resource Hub, co-led with UNDP.
- UN Development Programme (UNDP) — Digital Public Infrastructure guidance and co-stewardship of the Digital Public Goods Alliance.
- UN General Assembly — Global Digital Compact (2024) and associated digital cooperation follow-up processes.
- International Telecommunication Union (ITU) — GovStack Initiative (with GIZ, Estonia and UNDP), reusable government Building Block specifications.
- UN Conference on Trade and Development (UNCTAD) — Digital Economy Report series, covering cross-border data flows and data sovereignty for developing economies.
- World Bank Group — Identification for Development (ID4D) Practitioner's Guide and Principles on Identification for Sustainable Development.
- World Bank — "Resilient, Secure and Trusted: The Next Frontier for Digital Public Infrastructure" (2025).
- World Economic Forum — "Building Security into India's Digital Public Infrastructure" (October 2025) and DPI security risk taxonomy.

- OECD — Going Digital project and DPI security and governance guidance, including modular and federated architecture recommendations.

European Union, ENISA and European bodies

- European Union — General Data Protection Regulation (GDPR), Regulation (EU) 2016/679, including Chapter V cross-border transfer rules and Article 48.
- European Union — Cloud and AI Development Act (CADA), proposed 3 June 2026 as part of the European Technological Sovereignty Package.
- European Union — EU AI Act, in force August 2024; classifies AI used in biometric identification, social benefits determination and essential-services access as high risk.
- European Union — Digital Services Act (DSA) and Digital Markets Act (DMA), platform governance and gatekeeper obligations.
- European Union — Network and Information Security Directive 2 (NIS2), critical-infrastructure cybersecurity obligations across Member States.
- European Union Agency for Cybersecurity (ENISA) — cybersecurity guidance, threat landscape reporting and technical input to the EU Cybersecurity Certification Scheme for Cloud Services (EUCS).
- Court of Justice of the European Union — Schrems II (Case C-311/18, 2020), invalidating the EU–US Privacy Shield.
- European Data Protection Board — Recommendations 01/2020 on supplementary measures for cross-border data transfers.
- European Commission — European Technological Sovereignty Package (3 June 2026), including the EU Open Source Strategy and Chips Act 2.0.
- European Commission, DG CONNECT — Cloud Sovereignty Framework and “Shaping Europe’s Digital Future” portal.

United States: statutes, authorities and agencies

- Clarifying Lawful Overseas Use of Data Act (CLOUD Act), 2018 — establishes provider “possession, custody, or control”, not data location, as the trigger for compelled disclosure.
- Foreign Intelligence Surveillance Act (FISA) Section 702 — authorises US agencies to direct US-based providers to disclose communications of non-US persons abroad.
- Executive Order 12333 (1981, as amended) — foundational authority for US signals intelligence collection outside the United States.
- USA PATRIOT Act Section 215 — authority for compelled production of “tangible things” relevant to foreign intelligence investigations.
- United States v. Microsoft Corp. (2018) — the Dublin email-disclosure dispute that the CLOUD Act rendered moot.
- US National Institute of Standards and Technology (NIST) — Digital Identity Guidelines (SP 800-63-4); Secure Software Development Framework (SP 800-218); Cybersecurity Framework; AI Risk Management Framework.

- US Cybersecurity and Infrastructure Security Agency (CISA) — advisories, Known Exploited Vulnerabilities catalogue and critical-infrastructure guidance.
- US National Trade Estimate Report on Foreign Trade Barriers, 2026 edition — trade-policy treatment of foreign data localisation and sovereign cloud measures.

Other national and regional regimes

- China — Cybersecurity Law (2017), Data Security Law (2021), Personal Information Protection Law (2021), and Article 7 of the National Intelligence Law (2017).
- Russian Federation — Federal Law No. 152-FZ "On Personal Data", requiring localisation of citizens' personal data on servers within Russia.
- India — Digital Personal Data Protection Act (2023) and the Data Empowerment and Protection Architecture / Account Aggregator consent framework.
- Brazil — Lei Geral de Proteção de Dados Pessoais (LGPD), with extraterritorial effect and an independent supervisory authority.
- Estonia — Data Embassy concept and bilateral treaty with Luxembourg, extending Estonian jurisdiction to offshore infrastructure.
- Canada — Foreign Extraterritorial Measures Act and the emerging Canadian Sovereign Cloud Initiative.
- African Union — Digital Transformation Strategy for Africa (2020–2030) and the AfCFTA e-commerce protocol.
- Association of Southeast Asian Nations — ASEAN Digital Masterplan 2025.

Cloud and technology providers

- Microsoft — Azure sovereign cloud documentation: EU Data Boundary, Managed HSM external key storage, confidential computing.
- Google Cloud — Cloud External Key Manager and Cloud Key Management Service documentation on customer-managed and customer-held keys.
- Amazon Web Services — Key Management Service, Nitro Enclaves for confidential computing, and the AWS Digital Sovereignty Pledge.
- Thales CipherTrust — guidance on Bring Your Own Key, Hold Your Own Key and external key management for cloud data sovereignty.
- Utimaco — Enterprise Secure Key Manager, general-purpose HSMs and double key encryption reference architectures.
- Open Source Security Foundation / Linux Foundation — OpenSSF Scorecard and Supply-chain Levels for Software Artifacts (SLSA).

DPI and DPG frameworks and communities

- Digital Public Goods Alliance — DPG Registry, DPG Standard and the 50-in-5 campaign.
- MOSIP Foundation / IIIT-Bangalore — security architecture documentation for digital identity digital public goods.

- Mojaloop Foundation — security model and architecture for interoperable instant payment infrastructure.
- Estonia X-Road / Nordic Institute for Interoperability Solutions — secure government data exchange reference architecture.
- Centre for Digital Public Infrastructure / IIIT-Bangalore — DPI architecture and governance guidance, including G2P Connect specifications.
- Co-Develop Fund and the Global DPI Summit — country DPI journey funding and convening.
- Bill & Melinda Gates Foundation — Digital Public Infrastructure programme funding MOSIP, Mojaloop, DHIS2 and the Upanzi Network.
- Digital Impact Alliance — Principles for Digital Development.

Threat intelligence, standards and technical references

- CrowdStrike — 2026 Global Threat Report (Counter Adversary Operations).
- MITRE ATT&CK Framework — knowledge base of adversary tactics, techniques and procedures for nation-state threat modelling.
- OWASP Foundation — OWASP API Security Top 10.
- Center for Internet Security — CIS Benchmarks for cloud infrastructure and operating system hardening.
- ISO/IEC — ISO/IEC 27001 (Information Security Management) and ISO/IEC 42001 (AI Management Systems).
- Center for Strategic and International Studies — Significant Cyber Incidents Tracker and “Approaches to Digital Public Infrastructure in the Global South”.
- IBM — Cost of a Data Breach Report (2024 edition, regional analysis).
- Thomas Murray — GCC Regional Risk Update series (2026).
- Dubai Electronic Security Center — Dubai Cyber Security Strategy, cited as a reference model for institutionalised cyber governance.